**Date: 09/07/2025**

## Lab Practical #06:

Study Client-Server Socket programming - TCP & UDP

## Practical Assignment #06:

1. **Write a C/Java code for TCP Server-Client Socket Programming.**
2. **Write a C/Java code for UDP Server-Client Socket Programming.**

## 1. For TCP Server-Client:

### TCP Server Program:

```
import java.io.*;
import java.net.*;

public class TCPServer {
  public static void main(String[] args) throws IOException {
    ServerSocket serverSocket = new ServerSocket(3000);
    System.out.println("Server is running.");

    Socket clienSocket = serverSocket.accept();
    System.out.println("Client Connected.");

    BufferedReader in = new BufferedReader(new InputStreamReader(clienSocket.getInputStream()));
    PrintWriter out = new PrintWriter(clienSocket.getOutputStream(), true);

    String message = in.readLine();
    System.out.println("Client Says: " + message);

    out.println("Message recieved by the server.");

    clienSocket.close();
    serverSocket.close();
  }
}
```

### TCP Client Program:

```
import java.io.*;
import java.net.*;
```

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```
public class TCPClient {
    public static void main(String[] args) throws IOException {
        Socket socket = new Socket("localhost", 3000);

        PrintWriter out = new PrintWriter(socket.getOutputStream(), true);

        BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));

        out.println("Hello from client");

        String response = in.readLine();
        System.out.println("Server says: " + response);

        socket.close();
    }
}
```

## 2. For UDP Server-Client:

### UDP Server Program:

```
import java.io.*;
import java.net.*;

public class UDPServer {
    public static void main(String[] args) throws IOException {
        DatagramSocket ds = new DatagramSocket(3001);
        byte[] receive = new byte[65535];

        DatagramPacket DpReceive = null;

        while (true) {
            DpReceive = new DatagramPacket(receive, receive.length);
            ds.receive(DpReceive);

            System.out.println("Client:- " + data(receive));

            if (data(receive).toString().equals("bye")) {
                System.out.println("Client sent bye, Exiting.");
                break;
            }

            receive = new byte[65535];
        }

        ds.close();
    }
```

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```
   public static StringBuilder data(byte[] a) {
      if (a == null) return null;

      StringBuilder ret = new StringBuilder();
      int i = 0;

      while (a[i] != 0) {
         ret.append((char) a[i]);
         i++;
      }

      return ret;
   }
}
```

## UDP Client Program:

```
import java.io.*;

import java.net.*;

import java.util.*;


public class UDPClient {

   public static void main(String args[]) throws IOException {

      Scanner sc = new Scanner(System.in);


      DatagramSocket ds = new DatagramSocket();


      InetAddress ip = InetAddress.getLocalHost();

      byte buf[] = null;


      while (true) {

         String inp = sc.nextLine();
```

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```
            buf = inp.getBytes();

            DatagramPacket DpSend = new DatagramPacket(buf, buf.length, ip, 3001);

            ds.send(DpSend);

            if (inp.equals("bye")) break;
        }

        sc.close();
        ds.close();
    }
}
```